

Shibani Kumar

Dallas, TX | shibanik11@gmail.com | +1 619-648-0254 | linkedin.com/in/Shibani-Kumar11

Machine Learning Engineer with over 5 years of experience designing and deploying ML pipelines using Python, PyTorch, and Hugging Face. Improved model performance for SaaS clients and developed ETL frameworks that streamlined data workflows and supported client onboarding. Experienced in fine-tuning LLMs, RAG systems, and deploying high-throughput inference workflows. .

Education

The University of Texas at Dallas

Aug 2024 - Present

Master of Science in Information Technology and Management

GPA: 3.767

Coursework: Python, Advanced Statistics, Database Foundations, Business Analytics with R, Big Data, Predictive Analytics, Applied Machine Learning, Deep Learning, NLP, Applied AI, AWS Cloud Solutions Architecture

Recipient of Menka Hemrajani Fellowship, UTD Naveen Jindal School of Management

BNM Institute of Technology

Aug 2013 - Jun 2017

Bachelor of Engineering in Telecommunication Engineering

Technical Skills

Programming: Python, SQL, R, GitHub

ML/AI: HuggingFace Transformers, PyTorch, LangChain, Scikit-learn, TensorFlow, FastAPI, LLM Fine-Tuning, W&B

Data Engineering: ETL, Kafka, Apache Spark, PySpark, Hadoop, DynamoDB

Analytics/Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Google Analytics

Cloud/Deployment: Kubernetes, Flask, Streamlit, Docker, CI/CD, Azure, AWS

Research Experience

University of Texas, Dallas | Research Worker - LLM-Based AI System Development

May 2025 – Dec 2025

- Architected an end-to-end agentic LLM workflow using LangChain and LangGraph, orchestrating multi-step reasoning and tool-calling agents to automate scientific hypothesis generation, reducing manual research effort by 50%.
- Designed and optimized large-scale ingestion and evaluation pipelines for over 20,000 research papers, implementing structured extraction and custom benchmarking metrics to improve retrieval coverage by 40%.

Professional Experience

Moolya Software Technologies | Senior Machine Learning Engineer

Aug 2020 – Jun 2022

- Designed and deployed scalable ETL and feature engineering pipelines in Python, SQL, and Scikit-learn, ingesting multi-source SaaS data (Google Ads, Meta Ads, Spotify), improving model accuracy by 15% and reducing processing latency by 30%.
- Collaborated with cross-functional teams to design ML-driven analytics and created Tableau dashboards using Python facilitating detailed data analysis and performance modeling for onboarding clients worth 5M\$.
- Led technical mentorship for 3 junior engineers, standardizing pipeline modularization, SQL optimization, and model validation, which cut onboarding ramp-up time by 35% and boosted team productivity.

Accenture | Software Engineer - Machine Learning

Dec 2017 – Aug 2020

- Engineered distributed data processing pipelines using PySpark, SQL, and Python, automating large-scale order workflow processing for AT&T and reducing manual operational effort by 30%.
- Analyzed customer journey and product selection data using SQL, Python(Pandas, Scikit-learn) and Tableau, identifying bottlenecks in orders workflow and improving conversion tracking by 25%.

Academic Projects

AI-Powered Text Summarizer (HackUTD Project)

Nov 2024

- Developed an AI-powered text summarization tool using Meta Llama-3.1-8B - Instruct model to generate concise summaries of long-form text with an average 35% reduction in text and retaining 90% of key information.
- Built an interactive UI using Streamlit for seamless user interaction(100+users) to input text and instantly generate summaries, that improved reading experience by almost 40%
- Integrated Flask as backend to handle SambaNova API requests and process data with less than 2s latency per query.

Loan Default Prediction using Machine Learning

Oct 2025

- Built and optimized supervised ML models (Logistic Regression, Random Forest, XGBoost) to predict loan default risk on 67,000 borrower records with 35 financial and demographic features.
- Tuned hyperparameters using grid-search cross-validation and achieved 93% ROC-AUC and 0.89 F1-score with XGBoost after hyper-parameter optimization.

LLM-Based Content Moderation System Using Fine-Tuned Decoder Models

Nov 2025

- Built an end-to-end GenAI pipeline to mitigate toxic behaviour on social media (Reddit) using a system of pre-trained and fine-tuned decoder-based LLMs, achieving 85% true positive rate while improving label validity from 82% to 90% via constrained decoding.
- Deployed optimized models using vLLM for high-throughput inference, reducing latency by 4 times and implementing parameter-efficient adaptation strategies to enable scalable and cost-efficient production inference.